1（a)

select pid, people.name, city.name
from people join city using(cid)
where pid = 1

```
1  select pid, people.name, city.name
2  from people join city using(cid)
3  where pid = 1
4
```

	pid	name	name
1	1	uqxmpyncxlunyb	烟台

```
执行完成。
结果: 1 行返回，耗时 5ms
在行 1:
select pid, people.name, city.name
from people join city using(cid)
where pid = 1
```

1 (b)

select vaccine.name,count(*)test_times
from inoculate join vaccine using(vcid)
where pid = 1
group by vaccine.name

```
select vaccine.name,count(*)test_times
from inoculate join vaccine using(vcid)
where pid = 1
group by vaccine.name
```

	name	test_times
1	腺病毒载体疫苗	1

1（c)

select result , pcr_time
from pcrtest
where pid = 1
order by pcr_time DESC
limit 1

SQL 1

```
select result , pcr_time
from pcrtest
where pid = 1
order by pcr_time DESC
limit 1
```

	result	pcr_time
1	阴性	236

1（d)

select distinct city.name ,risk_level
from city
where city.cid in
(select cid
from visit join venue using(vid)
where pid = 1 and visit.leave_time + 100 > 1000
)

```sql
select distinct city.name ,risk_level
from city
where city.cid in
(select cid
from visit join venue using(vid)
where pid = 1 and  visit.leave_time + 100 > 1000
)
```

	name	risk_level
1	烟台	低风险

2

select pid, name, test_times
from people left outer join
(select count(*)test_times, pid
from pcrtest
where pcr_time > 900
group by pid
) using(pid)
order by test_times DESC

```sql
select pid, name, test_times
from people left outer join
(select count(*)test_times, pid
from pcrtest
where pcr_time > 900
group by pid
) using(pid)
order by test_times DESC
```

	pid	name	test_times
1	1705	mqtnvkofcutznc	2
2	115	bsumpizqfs	1

执行完成。
结果：10000 行返回，耗时 16ms
在行 1:

3

select pid, people.name, max(visit.leave_time)
from people join visit using(pid) join venue using(vid)
where people.cid = 2 and venue.cid = 1 and visit.leave_time+100 > 1000
group by pid

```
select pid, people.name, max(visit.leave_time)
from people join visit using(pid) join venue using(vid)
where people.cid  = 2 and venue.cid = 1 and visit.leave_time+100 > 1000
group by pid
```

	pid	name	max(visit.leave_time)
1	2166	zunnyvwucjchh	938
2	2630	dcyzehyxgfwwa	926
3	7397	oiyjaeqllzycp	934
4	8022	pahogwdzbru	922

执行完成。
结果：4 行返回，耗时 29ms

4.

select pid, people.name, vaccine.name, count(*)vc_times
from people join inoculate using(pid) join vaccine using(vcid)
group by pid, people.name, vaccine.name
having vc_times = required_doses

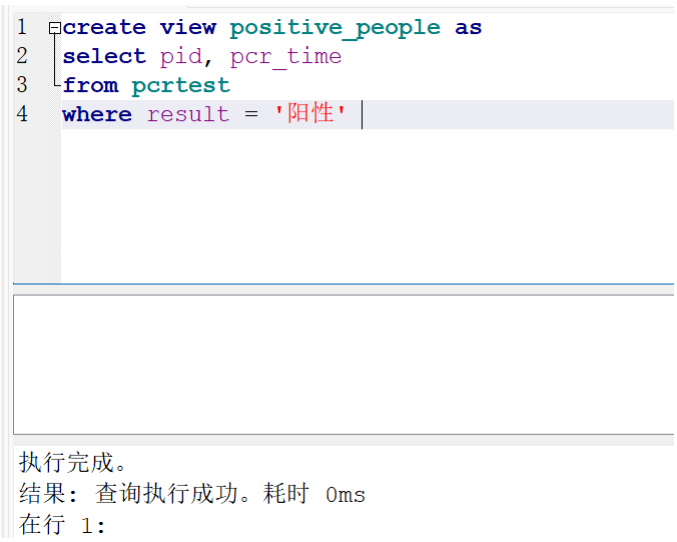
```
select pid, people.name, vaccine.name, count(*)vc_times
from people join inoculate using(pid) join vaccine using(vcid)
group by pid, people.name, vaccine.name
having vc_times = required_doses
```

	pid	name	name	vc_times
1	0	awvtmfkzkrzhf	灭活疫苗	2
2	1	uqxmpyncxlunyb	腺病毒载体疫苗	1
3	2	szaqtuadhxff	灭活疫苗	2
4	3	qmtkklvgfsfeg	灭活疫苗	2
5	4	opogbxbbmkvc	重组亚单位疫苗	3

执行完成。
结果：9269 行返回，耗时 54ms
在行 1:
select pid, people.name, vaccine.name, count(*)vc_times

5

```
create view positive_people as
select pid, pcr_time
from pcrtest
where result = '阳性'
```



6.

```
update people
set status ='红码'
where pid in(
select pid
from (
select pid, max(pcr_time)pcr_time,result
from pcrtest
where pid in(select pid from positive_people)
group by pid
)join people using(pid)
where result = '阳性')
```

```sql
update people
set status ='红码'
where pid in(
select pid
from (
select pid, max(pcr_time)pcr_time,result
from pcrtest
where pid in(select pid from positive_people)
group by pid
)join people using(pid)
where result = '阳性')
```

执行完成。
结果：查询执行成功。耗时 4ms，8 行数据受影响
在行 1:
update people

7.

select people.name, venue.name, enter_time, leave_time, pcr_time
from people join visit using(pid) join venue using(vid) ,positive_people
where people.pid = positive_people.pid
and pcr_time-14 < leave_time and pcr_time > enter_time

```sql
select people.name, venue.name, enter_time, leave_time, pcr_time
from people join visit using(pid) join venue using(vid) ,positive_people
where people.pid = positive_people.pid
and pcr_time-14 < leave_time and pcr_time > enter_time
```

	name	name	enter_time	leave_time	pcr_time
1	twwngpmqyhn	alscsbcjxc	133	152	161
2	ortqzrzkcmdg	ehcszadbkp	44	93	60

执行完成。
结果：21 行返回，耗时 65ms

8.

update people
set status ='黄码'
where status ='绿码'and pid in(
select pid
from (select venue.vid,pcr_time, enter_time pe_time, leave_time pl_time
from people join visit using(pid) join venue using(vid) ,positive_people

where people.pid = positive_people.pid
and pcr_time-14 < leave_time and pcr_time > enter_time) NATURAL join visit
where(pe_time < pcr_time-14 and pl_time <pcr_time and ((enter_time > pcr_time-14 and enter_time < pl_time)or(leave_time > pcr_time-14 and leave_time <pl_time)))
or(pcr_time - 14 > pe_time and pcr_time < pl_time and ((enter_time>pcr_time-14 and enter_time <pcr_time)or (leave_time > pcr_time-14 and leave_time < pcr_time)))
or(pe_time > pcr_time-14 and pcr_time < pl_time and ((enter_time > pe_time and enter_time < pcr_time)or(leave_time > pe_time and leave_time < pcr_time)))
or (pe_time > pcr_time-14 and pl_time < pcr_time and((enter_time>pe_time and enter_time < pl_time)or (leave_time >pe_time and leave_time < pl_time)))
)

```
update people
set status ='黄码'
where status ='绿码'and pid in(
select pid
from (select  venue.vid,pcr_time, enter_time pe_time, leave_time pl_time
from people join visit using(pid) join venue using(vid) ,positive_people
where people.pid = positive_people.pid
and pcr_time-14 < leave_time and pcr_time > enter_time) NATURAL join visit
where(pe_time < pcr_time-14 and pl_time <pcr_time and ((enter_time > pcr_time-14 and enter_time < pl_time)or(leave_time > pcr_time-14 and leave_time <pl_time)))
or(pcr_time - 14 > pe_time and pcr_time < pl_time and ((enter_time>pcr_time-14 and enter_time <pcr_time)or (leave_time > pcr_time-14 and leave_time < pcr_time ))
or(pe_time > pcr_time-14 and pcr_time < pl_time and ((enter_time > pe_time and enter_time < pcr_time)or(leave_time > pe_time and leave_time < pcr_time)))
or (pe_time > pcr_time-14 and pl_time < pcr_time and((enter_time>pe_time and enter_time < pl_time)or (leave_time >pe_time and  leave_time < pl_time)))
)
```

```
执行完成。
结果：查询执行成功。耗时 223ms，9 行数据受影响
在行 1:
update people
set status ='黄码'
```